

Online Appendix: Emoji Substitution and Appended Discourse Markers in NLI

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This appendix accompanies the paper. Section numbers match the pointers in the main text (App. S1 to S13). Unless stated otherwise, results are from one training run (seed 42), accuracy is scored against inherited labels, five-encoder means exclude DeBERTa-v3-base, and losses are negative in tables.

S1 Emoji Inversion, the Normalizer, and Evaluation Noise

This section separates two operations that map emoji back to nouns. It covers the five encoders and three splits for which both operations were checked, at seed 42. DeBERTa-v3-base was not part of this check; its normalization results are in Section S3.

Exact inversion. Emoji substitution can be undone exactly. The map is one-to-one, and every replaced span is stored with its pair. Restoring only the stored spans reproduces the paired clean text and the paired clean token identifiers for every pair, in all 15 combinations of encoder and split. The emoji sets contain 6,430 SNLI pairs, 522 MultiNLI matched pairs, and 574 MultiNLI mismatched pairs. Tokenization used a maximum length of 128.

The inference-time normalizer. Emoji normalization does not use the stored spans. It replaces every occurrence of each of the 16 mapped emoji with its noun, longest emoji string first, and then collapses irregular whitespace. It does not remove markers. It differs from the exact inverse in two ways. First, it writes each noun in lower case, so a capitalized source noun such as *Dog* returns as *dog*. Second, the whitespace step also changes irregular spacing that was present in the source text. No emoji that occur naturally in the source text are involved.

The normalizer alters 328 SNLI pairs, 148 MultiNLI matched pairs, and 200 MultiNLI mismatched pairs relative to paired clean. These counts are the same for every encoder, because they depend on the text only. Every altered sentence differs from its source only in case or whitespace. On SNLI, 324 sentences differ in case only and 19 in whitespace only. On MultiNLI matched, 36 differ in case only, 128 in whitespace only, and 7 in both. On MultiNLI mismatched the counts are 57, 165, and 8.

Token identifiers and logits. Whether the altered text changes the model input depends on the tokenizer (Table S1). ELECTRA-small is uncased, so its token

Table S1. Emoji normalization compared with paired clean input for the clean baseline (seed 42). Pairs per split: 6,430 SNLI, 522 MultiNLI matched (m), 574 mismatched (mm). Text and Tokens: pairs whose normalized text or token identifiers differ from paired clean. Logits: pairs with any differing logit. Max: largest absolute logit difference. Pred.: pairs with a different prediction. Resid.: normalized accuracy minus paired-clean accuracy (pp).

Encoder	Split	Text	Tokens	Logits	Max	Pred.	Resid.
ELECTRA-small	SNLI	328	0	0	0	0	+0.00
ELECTRA-small	m	148	0	193	0.0508	0	+0.00
ELECTRA-small	mm	200	0	105	0.0781	1	−0.17
RoBERTa-base	SNLI	328	328	317	1.64	3	+0.02
RoBERTa-base	m	148	148	150	0.426	4	+0.38
RoBERTa-base	mm	200	200	194	1.03	4	+0.00
RoBERTa-large	SNLI	328	328	319	1.72	2	−0.03
RoBERTa-large	m	148	148	147	0.699	2	−0.38
RoBERTa-large	mm	200	200	197	3.27	1	−0.17
TimeLM-21	SNLI	328	309	1,884	1.44	3	+0.05
TimeLM-21	m	148	29	208	0.711	1	−0.19
TimeLM-21	mm	200	50	251	2.36	2	+0.35
BERTweet	SNLI	328	309	1,814	0.441	2	+0.03
BERTweet	m	148	29	233	0.877	1	−0.19
BERTweet	mm	200	50	244	0.719	2	−0.35

identifiers are unchanged for all pairs. The two RoBERTa tokenizers change identifiers for every altered pair. The TimeLM-21 and BERTweet tokenizers absorb part of the whitespace differences and change identifiers for 309, 29, and 50 pairs. The logits of the normalized input equal those of the paired clean input in 1 of the 15 cells, ELECTRA-small on SNLI.

Prediction identity. Predictions are almost always unchanged. Normalized and paired clean input receive the same prediction for 99.23% to 100% of pairs, and 0 to 4 predictions differ per cell. Normalized accuracy therefore lies within 0.38 points of paired-clean accuracy in every cell, in either direction. The per-encoder means of this residual range from −0.20 to +0.13 points.

Evaluation noise on identical text. Residuals of this size are at the level of evaluation noise. Two observations support this. First, ELECTRA-small shows logit differences on both MultiNLI splits although its token identifiers are identical there, so these differences cannot arise from the text. Second, the evaluation sets contain a condition whose text is identical to paired clean for all 7,526 emoji pairs. The accuracy of the clean baseline on this condition differs from its paired-clean accuracy in 5 of the 15 cells: by −0.17 points for ELECTRA-small on MultiNLI mismatched, +0.02 for TimeLM-21 on SNLI, +0.35 for TimeLM-21 on MultiNLI mismatched, +0.02 for BERTweet on SNLI, and −0.17 for BERTweet on MultiNLI mismatched. One pair corresponds to 0.19 points on MultiNLI

Table S2. Judgments in the label-preservation review. Wilson 95% intervals are given for the majority rate. Unanimity and individual judgments are descriptive, since the same items are rated three times.

Unit	Preserved	Total	Rate (%)	Wilson 95% interval
Items, majority	240	240	100.00	98.42–100
Items, unanimous	239	240	99.58	–
Individual judgments	719	720	99.86	–
Each dataset, majority	120	120	100.00	96.90–100
Each condition, majority	60	60	100.00	93.98–100
Each label, majority	80	80	100.00	95.42–100

matched and 0.17 points on mismatched. The cause of these differences was not identified. Differences below about 0.4 points should not be interpreted.

S2 Label-Preservation Review

The review checks whether edited pairs keep the label of their source. It is a verification in which reviewers saw that label. It is not an independent annotation of the edited pairs.

Sampling design. The sample has 240 unique items from the development data: the SNLI validation set and the MultiNLI development holdout. It is balanced over two datasets, four conditions, and three labels, with ten items per stratum. The four conditions are emoji substitution, fold-1 markers seen in augmentation, fold-1 held-out markers, and the combined edit. Markers in the two marker conditions are appended at the end of the hypothesis. Items were selected without reference to model outputs or model correctness, and no item was removed after selection.

What reviewers saw and the question asked. Each item showed the original premise and hypothesis, the inherited label, the edited premise and hypothesis, and the type of edit. Model predictions were not shown. The question was whether the displayed label remains correct after the edit. Reviewers answered yes or no and gave a written reason for each no. All three reviewers completed all 240 items, in the same randomized order.

Aggregate judgments. All 240 items were judged label-preserving by at least two of three reviewers, and 239 by all three (Table S2). The Wilson 95% interval for the majority rate is 98.42–100%. The three reviewers gave 239, 240, and 240 yes judgments. One response contained a typing error and was read as yes. The majority result does not depend on that response or on the one dissent.

The one dissent. The dissent concerns an SNLI entailment item in the seen-marker condition. The premise is “A man sitting at a desk, repairing antique

Table S3. Reviewed items per marker. The seen and held-out conditions append the marker to the hypothesis. The combined condition also applies emoji substitution and uses the fold-1 held-out markers. The 60 emoji items carry no marker.

Condition	<i>fr</i>	<i>tbh</i>	<i>ngl</i>	<i>real talk</i>	<i>on god</i>	<i>istg</i>	<i>frfr</i>	<i>deadass</i>	<i>no cap</i>	Total
Seen markers	9	9	15	6	13	8	–	–	–	60
Held-out markers	–	–	–	–	–	–	24	12	24	60
Combined	–	–	–	–	–	–	12	28	20	60

typewriters.” The hypothesis “a man sits” became “a man sits *on god*”. One reviewer judged that *on god* could attach literally to *sits*. The other two read it as the intended discourse marker and judged the label preserved.

Items per marker. All nine markers occur in the sample (Table S3). In the two marker conditions each marker has 6 to 24 items. The sample was balanced by condition and label, not by marker, so the counts are unequal. In the combined condition the marker stands at the end of the hypothesis in 18 items, at its start in 23, and at the end of the premise in 19.

What the review does not support. The review supports the use of inherited labels for the four reviewed conditions, in the sense checked by reviewers who saw the label. It does not support five further conclusions. First, it provides no independent labels and no measure of agreement between annotators. Second, it does not cover the formal phrases, the other appended words, the component words, the content-word phrases, or the phrases appended to SICK and ANLI. Third, it does not cover markers placed at the start of the hypothesis or the end of the premise outside the combined condition. Fourth, it does not check the original labels of SNLI and MultiNLI. Fifth, it does not support conclusions about a single marker: the counts per marker are small, and the design gives no estimate of how often a literal reading arises. The review was stratified by dataset, condition, and label. Its 120 emoji items nevertheless cover all five replacement structures: 40 items replace one noun in both sentences, 38 in the premise only, 20 in the hypothesis only, 14 replace two nouns in the same sentence, and 8 replace two nouns, one per sentence.

S3 Training Setup and DeBERTa-v3-base Results

Shared training setup. All six encoders follow one training protocol. One three-label classifier is trained per encoder and dataset, and the MultiNLI classifier is evaluated on both validation splits. Training runs for three epochs with the default AdamW optimizer, a linear schedule, warmup ratio 0.06, zero weight decay, and maximum length 128. The model is evaluated and saved after each epoch. We keep the checkpoint with the highest clean development accuracy, and the first such checkpoint in case of a tie. All results in this section use seed 42.

Table S4. Encoder-specific training settings. LR: learning rate. Acc.: gradient accumulation steps.

Encoder	LR	Batch	Acc.	Tokenizer settings
ELECTRA-small	2e-5	32	1	default
RoBERTa-base	2e-5	32	1	default
RoBERTa-large	1e-5	16	2	default
TimeLM-21	2e-5	32	1	model-card preprocessing of user names and URLs
BERTweet	2e-5	32	1	slow tokenizer, normalization disabled
DeBERTa-v3-base	2e-5	32	1	fast tokenizer, SentencePiece byte fallback enabled

Table S4 lists the settings that differ between encoders. The effective batch size is 32 for every encoder.

DeBERTa-v3-base setup. The hyperparameters of DeBERTa-v3-base are those of RoBERTa-base. Training takes 51,504 optimizer steps on SNLI and 34,977 on MultiNLI. The weights are held in 32-bit floating point, which is verified when the model is loaded, and training uses bf16 autocast. Fine-tuning from half-precision weights did not converge. RoBERTa-large is also trained with bf16. Development accuracy after the three epochs was 92.45%, 93.18%, and 93.22% on SNLI, and 89.71%, 90.18%, and 90.15% on MultiNLI. We train only the clean baseline. Emoji normalization is applied to this baseline at inference.

SentencePiece byte fallback. With its default tokenizer, DeBERTa-v3-base cannot distinguish 5 of the 16 emoji. The emoji for *person*, *child*, *cat*, *horse*, and *bicycle* all map to the same unknown-token identifier, and 36 of 194 emoji spans in a test sample contain the unknown token. The SentencePiece model of the checkpoint was trained with byte fallback, but the converted fast tokenizer has it disabled. We enable it. All 16 emoji then receive distinct encodings, alone and in context, and no emoji span contains the unknown token. The resulting tokenizer agrees with the SentencePiece model on all 300 evaluation texts compared; the default tokenizer disagrees on 30. The setting is applied each time the tokenizer is loaded, since a reload with default settings drops it. On text without emoji, both tokenizers give identical token identifiers. We do not attribute the smaller emoji loss of DeBERTa-v3-base to the tokenizer, because architecture and pretraining differ as well.

Per-cell results. Both edits lower the paired accuracy of DeBERTa-v3-base on every split (Table S5). Clean accuracy on the full splits is 92.49% on SNLI test, 89.84% on MultiNLI matched, and 90.34% on mismatched. Emoji substitution lowers paired accuracy by 1.63, 2.68, and 3.31 points, with Holm-adjusted $p < .001$, $p = .038$, and $p = .002$. Emoji normalization raises accuracy on emoji input by 1.62, 2.68, and 3.31 points ($p < .001$, $p = .037$, $p = .002$). Normalized accuracy

Table S5. DeBERTa-v3-base per-cell results against inherited labels (seed 42, one training run). Accuracy in %, changes in pp. Emoji: n pairs, paired-clean accuracy, accuracy on emoji input, emoji loss, accuracy after emoji normalization, and repair. Held-out markers: marker loss per fold and its mean. [†]Not significant after Holm correction; all other changes are significant within the family of the split.

Split	Emoji						Held-out markers			
	n	Clean	Emoji	Loss	Norm.	Repair	Fold 1	Fold 2	Fold 3	Mean
SNLI	6,430	92.50	90.87	-1.63	92.49	+1.62	-13.11	-1.90	-6.71	-7.24
MultiNLI m	522	88.89	86.21	-2.68	88.89	+2.68	-8.20	-0.50	-3.44	-4.05
MultiNLI mm	574	87.46	84.15	-3.31	87.46	+3.31	-6.31	-0.21 [†]	-3.79	-3.44
Mean				-2.54			-9.21			-4.91

is 0.02 points below paired-clean accuracy on SNLI and equal to it on both MultiNLI splits. Held-out markers lower accuracy in every fold and split, by 4.91 points on average. Eight of the nine fold-level losses are significant. The exception is fold 2 on MultiNLI mismatched (-0.21 pp, $p = .133$). These values form a sixth, separate row and are not included in five-encoder means. The tests belong to a separate family (Section S4).

S4 Statistical Test Families

Each group of tests forms its own family (Table S6). All tests compare two conditions or two systems on the same source pairs. All results are from seed 42, except the non-inferiority test, which uses five training runs.

The four specified contrasts. The four contrasts of one encoder, split, and fold form one family. The emoji loss does not depend on the fold, so it is tested once in each of the three fold families of a cell. The adjusted p values differ slightly between folds, but no decision changes. Counts of significant emoji cells use the fold-1 adjustment and count each encoder and split once.

DeBERTa-v3-base. DeBERTa-v3-base is tested in a separate family of eleven exact tests per split. Of these, 9, 9, and 8 are significant on SNLI, MultiNLI matched, and MultiNLI mismatched. Applying the correction to continuity-corrected chi-square p values instead changes none of the 33 decisions. Effect sizes are comparable across the six encoders. Significance counts are not, and they are not added to the counts of the five encoders.

Crossed study. The 132 tests of the crossed study and their predicted directions were fixed before the predictions were computed, and 129 are significant. The phrase *on god* was chosen after its effect had been seen on the same evaluation data. These tests therefore describe that effect and do not confirm it independently. The comparison of observed with expected all-five failure counts is not part of this family and is exploratory.

Preregistered families. For the content-word phrases and for SICK and ANLI, predictions and decision rules were written before any edited input was built. The predictions required Holm-adjusted $p < .01$ together with a ratio of failure rates: at least 2 for the content-word phrases and at least 3 for SICK and ANLI. No correction is applied across the families of different encoders or splits.

Analyses without a family. The remaining analyses are exploratory and carry no multiplicity correction. They include the repair by emoji normalization, the flip rates, the analyses by replacement structure and noun, the content-word coding, and the ChaosNLI check. The non-inferiority test applies its decision rule to each cell separately, so a joint statement over the nine cells has no familywise control.

S5 Sensitivity to the Training Run and Truncation

Additional runs. The additional runs repeat the four specified contrasts with other seeds. ELECTRA-small is trained with seeds 13, 21, 87, and 101 on both datasets, in all three folds, and for all systems. RoBERTa-base, RoBERTa-large, TimeLM-21, and BERTweet are trained with seeds 13 and 87 on both datasets in fold 1 only. This gives 156 runs for ELECTRA-small and 24 for each of the other four encoders, 252 in total. Of these, 216 produce results and 36 compute statistics only. Together with seed 42, each ELECTRA-small cell has five runs and each cell of the other four encoders has three. DeBERTa-v3-base has one run and is not covered.

Cells. A cell is one combination of encoder, split, fold, and contrast. There are 78 cells: 30 for ELECTRA-small and 12 for each of the other four encoders.

Direction. The direction of every contrast recurs in every run (Table S7). In all 78 cells, every run shows a loss for the two edits and a gain for the two fix contrasts. In no cell does the seed-42 value differ in sign from the mean of the additional runs. A two-sided 95% Student- t interval across runs excludes zero in 74 of 78 cells. The four exceptions are fold-1 MultiNLI cells with three runs: normalization versus augmentation for RoBERTa-large on the matched split (+3.32 pp, interval -1.45 to $+8.09$) and the mismatched split (+2.61, -2.49 to $+7.72$), and for BERTweet on the mismatched split (+2.26, -0.03 to $+4.55$); and hybrid versus baseline for BERTweet on the mismatched split (+3.14, -1.38 to $+7.65$). Intervals from three runs have two degrees of freedom and are wide.

Magnitude. The size of the contrasts is less stable than their direction. The seed-42 value lies outside the range of the additional runs in 39 of 78 cells. By encoder, the counts are 9 of 30 for ELECTRA-small, 9 of 12 for RoBERTa-base, 6 of 12 for RoBERTa-large, 6 of 12 for TimeLM-21, and 9 of 12 for BERTweet. The largest difference between the seed-42 value and the mean of the additional runs is 10.70 points. It occurs for the marker loss of RoBERTa-base on SNLI in fold 1, which is -15.33 pp at seed 42 and -26.03 pp on average in the two

additional runs. For the four encoders with two additional runs, this criterion is weak. In some cells both additional runs give the same value, and any other seed-42 value then counts as outside the range.

Repeated runs, not exact seed replicates. The training code created the classification head before it applied the seed. A run with a given seed is therefore not reproducible bit for bit from that seed. We treat the additional runs as repeated training runs. They do not cover the control phrases, the gain from marker augmentation on held-out markers, or the crossed study.

Truncation. Truncation at the maximum length of 128 tokens explains neither loss. Emoji substitution pushes no input over 128 tokens in any of the 15 model-split cells of the five encoders. Some MultiNLI emoji pairs exceed 128 tokens before and after the edit: 6 of 522 matched pairs and 4 of 574 mismatched pairs for four encoders, and 6 and 6 for BERTweet. No SNLI emoji pair does. A held-out marker pushes at most 0.0305% of the pairs of a cell over 128 tokens, and at most 47 of 9,832 marked inputs exceed 128 tokens. No input exceeds 256 tokens. Restricting each comparison to pairs within the limit before and after the edit changes the paired accuracy change by at most 0.15 points. The diagnostics use the seed-42 clean baselines and cover emoji substitution and held-out markers at the three positions. They do not cover the control phrases, the combined condition, the crossed study, or DeBERTa-v3-base.

S6 Emoji Loss by Replacement Structure and by Noun

All values in this section come from the seed-42 clean baselines and are scored against inherited labels. The analyses are exploratory. DeBERTa-v3-base is a separate column and is not included in five-encoder means. The tables abbreviate the encoders: EL is ELECTRA-small, RB is RoBERTa-base, RL is RoBERTa-large, TL is TimeLM-21, BT is BERTweet, and DeB is DeBERTa-v3-base.

Replacement structure on SNLI. Most encoders order the five replacement structures as the five-encoder mean of the main text does (Table S8). Every encoder loses least when the same noun is replaced in both sentences, and every encoder loses most when two nouns are replaced, one per sentence. For BERTweet, replacement in the hypothesis only costs slightly less than replacement in the premise only. For DeBERTa-v3-base, two nouns in the same sentence cost less than one noun in the hypothesis only. The flip rates of DeBERTa-v3-base for the five structures, in the order of the table, are 1.70%, 3.35%, 7.55%, 4.92%, and 11.86%.

Replacement structure on MultiNLI. On both MultiNLI splits, replacement in the hypothesis only costs the five encoders more than the other two one-noun structures (Table S9). The five-encoder means for the premise only and for both sentences are close to each other on each split. Single cells vary widely because the subsets hold 122 to 254 pairs. The two-noun structures have 7 to 44 pairs per MultiNLI split. These are too few to report.

Replaced noun. Table S10 gives the loss for each of the 16 nouns. It uses the 4,640 SNLI test pairs with one replaced noun, so no pair is counted under two nouns. The seven nouns with at least 197 pairs have five-encoder means between -1.74 and -3.86 pp. The other nine nouns have 5 to 77 pairs, and their means range from $+2.11$ to -23.75 pp. Estimates for nouns with few pairs are unstable and should not be interpreted singly. Encoders can also disagree on a frequent noun. For *person*, the change ranges from 0.00 pp for BERTweet to -10.00 pp for TimeLM-21. Among pairs that replace the same noun in both sentences, the seven nouns with at least 123 pairs have five-encoder means between $+0.28$ pp (*person*) and -1.52 pp (*girl*).

Permutation tests. For each encoder we test whether the mean paired accuracy change is equal across the levels of a factor. The statistic is the between-level sum of squares weighted by level size, with 2,000 permutations on SNLI test. The smallest attainable p is 0.0005, and the p values are unadjusted (Table S11). Equality across the five structures is rejected at the smallest attainable p for all six encoders. Equality across nouns is rejected among pairs with one replaced noun, which mix the three one-noun structures. Among pairs that replace the same noun in both sentences, the loss differs detectably between nouns only for RoBERTa-base.

Adjusted model. A logistic model with standard errors clustered by source pair gives a consistent result for the five encoders on SNLI test. It models whether a clean-correct decision becomes a failure under emoji substitution (29,261 decisions from 6,190 source pairs). It adjusts for premise and hypothesis length, lexical overlap, label, number of replacements, noun category, and encoder. Its levels describe which sentences carry replacements, so they do not coincide with the five replacement structures. At two replacements and median overlap, the adjusted probability of a failure is 1.81% when both sentences carry the same replaced noun. It is 12.40% for replacement in the premise only, 22.50% when the two sentences carry different nouns, and 31.48% for replacement in the hypothesis only. The last estimate exceeds the first by 29.68 points (95% interval 23.55 to 35.80), but it rests on 31 source pairs and 144 decisions. The model was not fitted to MultiNLI.

Representation controls. Two controls on the seed-42 clean baselines describe how the emoji are encoded. For ELECTRA-small, whose vocabulary has no entry for the 16 emoji, the logits on emoji input equal those obtained with the unknown token at the same positions. This holds on all three splits, and no prediction differs. For RoBERTa-base and RoBERTa-large, we replaced each emoji with a string of the same byte length, keeping the encoded length of the pair. These length-matched strings lower accuracy more than the emoji do, by 3.530 to 9.195 points in the six model-split cells. Encoded length alone therefore does not account for the emoji loss of these two encoders. The length-matched strings also break the match of a repeated word. Where the same noun is replaced in both sentences, the two occurrences receive different strings in 2,922 of 2,934 SNLI

pairs, 149 of 152 MultiNLI matched pairs, and 138 of 141 mismatched pairs. The comparison therefore does not isolate one cause. TimeLM-21, BERTweet, and DeBERTa-v3-base were not tested.

Caveats. Four caveats apply to the analyses by structure and by noun. First, the structures are not crossed with source pairs. Each structure occurs in different pairs, so differences between source sentences are mixed with the effect of the structure. Second, the two-noun structures carry two replacements and the one-sided one-noun structures carry one, so structure is partly confounded with the number of replacements. Third, at most two occurrences are replaced per pair. A pair with two replaced nouns can therefore keep a further occurrence of one of these nouns as a word, and the structure is defined by the recorded replacements only. Fourth, all values come from one training run.

S7 Fold-Study Controls and Marker Position

Construction of the controls. In the fold study, every marker condition has two control conditions on the same source pairs. The formal phrase and the other appended word of a source pair are drawn, in a fixed rotation, from the phrases with the same word count as its marker. They are placed at the same position. The formal phrases are *honestly* and *seriously* (one word) and *in fact* (two words). The other appended words are *nearby* and *outside* (one word) and *in the, with it, and at this* (two words). The controls are matched on word count and position only. They are not matched on subword length, function, or frequency, and they were not reviewed for label preservation. All values in this section come from the seed-42 clean baselines and are scored against inherited labels.

Mean losses. The controls lower accuracy by about half as much as the held-out markers. With held-out markers at the end of the hypothesis, the five encoders give 45 model-split-fold cells. Over these cells, formal phrases lower accuracy by 5.06 points on average (range 0.98 to 15.04) and other appended words by 5.81 points (range 1.25 to 17.86). Held-out markers lower it by 10.13 points.

Marker minus control. Markers exceed the controls on average, but not in every fold (Table S12). The marker lowers accuracy more than the formal phrase in 35 of 45 cells and more than the other appended word in 31 of 45. A control is more damaging than the marker in the remaining 10 and 14 cells. All 10 cells for formal phrases are in fold 2, as are 13 of the 14 for other appended words. The remaining cell is RoBERTa-large on SNLI in fold 3 (+2.30 pp). The held-out markers of fold 2 are the one-word abbreviations *fr*, *tbh*, and *ngl*. Its controls are therefore *honestly* or *seriously* and the locatives *nearby* or *outside*, which can add a location to the hypothesis. Fold and control phrase are thus confounded. By encoder, the mean difference from formal phrases ranges from −1.50 pp (BERTweet) to −8.44 pp (RoBERTa-base). The mean difference from other appended words ranges from −1.49 pp (ELECTRA-small) to −8.45 pp (RoBERTa-base).

DeBERTa-v3-base. DeBERTa-v3-base shows the same dependence on the fold, in a separate test family. In fold 2, both controls lower its accuracy more than the markers on every split. The marker minus the formal phrase is +10.61, +4.64, and +5.35 pp on SNLI, MultiNLI matched, and MultiNLI mismatched. The marker minus the other appended word is +10.15, +8.17, and +7.51 pp. All six differences are Holm-significant. In fold 1, the marker is more damaging than the control in all six cells (−0.75 to −3.78 pp), and five are significant. In fold 3, the differences lie between −0.74 and +0.46 pp, and only one is significant (MultiNLI mismatched, formal phrase, −0.74 pp). For this encoder, markers exceed their controls only in fold 1.

Position. The marker loss is largest at the end of the hypothesis and smallest at the end of the premise (Table S13). Over the 45 cells of the five encoders, held-out markers lower accuracy by 10.13 points at the end of the hypothesis (range 1.12 to 22.32) and by 5.52 points at its start (range 0.05 to 14.78). Accuracy falls in all 45 cells at both positions. At the end of the premise the mean loss is 0.75 points, the change ranges from −8.89 to +0.11 pp, and accuracy falls in 38 of 45 cells. The loss at the end of the premise is near zero for every encoder except ELECTRA-small. These values are descriptive and unadjusted. Values by position for DeBERTa-v3-base are not reported.

S8 The Crossed Study

The crossed study classifies every source pair of the three splits under 24 conditions: clean, nine markers, six component words, three formal phrases, and five other appended words. Every phrase is appended at the end of the hypothesis. All values come from one training run per encoder (seed 42).

Table S14 lists the 23 non-clean conditions. Accuracy is scored against inherited labels. The flip rate needs no label and measures sensitivity, not error. DeBERTa-v3-base is not part of the five-encoder means. None of the 23 conditions was reviewed for label preservation, and the component words are synthetic ablations, not natural usage.

Component-word contrasts. A two-word marker usually causes more entailment failures than either of its component words. Each of *on god*, *real talk*, and *no cap* is compared with each of its two words, for each of the five encoders on each split, on the clean-correct entailments of that encoder. This gives 90 contrasts, tested with the two-sided exact McNemar test. The marker causes more failures in 79 contrasts and fewer in 9, and 2 are not significant (Table S15).

The nine contrasts in which the component word causes more failures are concentrated in two encoders. For TimeLM-21, *no* alone causes more failures than *no cap* on all three splits. For ELECTRA-small on both MultiNLI splits, *god* alone causes more failures than *on god*.

The 90 contrasts belong to one family of 132 tests under a single Holm correction at $\alpha = .05$, fixed before the predictions were produced. The family

also contains 30 contrasts on the share of clean-correct entailments moved to neutral (*on god* against *on* and against *god*) and 12 contrasts on all-five failure (*on god* against each two-word control, per split). In total, 129 of the 132 tests are Holm-significant. *On god* was chosen after its effect was seen in the fold study on the same evaluation data, so these tests describe the effect and do not confirm it independently. DeBERTa-v3-base is not part of this family.

Agreement between flip rate and accuracy loss. The ranking of conditions by flip rate is close to the ranking by accuracy loss. Across the 23 conditions, the five-encoder mean flip rate and mean accuracy loss have Spearman $\rho = 0.987$. The same four markers lead both rankings in the same order. The rankings are not identical. Of the 253 pairs of conditions, 8 are in inverted order: *fr* and *in fact*, *fr* and *with it*, *istg* and *in fact*, *istg* and *in the*, *frfr* and *god*, *talk* and *outside*, *seriously* and *outside*, and *in fact* and *in the*. For example, *frfr* lowers accuracy more than *god* (7.53 against 7.45 points), but *god* changes more predictions (11.6% against 11.2%).

Matching of the controls. The formal phrases and the other appended words are matched to the markers on word count and position only. They are not matched on subword length, discourse function, frequency, or semantic contribution. The locatives *nearby* and *outside* can add a location to the hypothesis. With them, the eight controls lower accuracy by 1.30 to 8.78 points for the five encoders. Without them, the range is 1.30 to 6.69 points.

S9 Content-Word Phrases and Content-Word Coding

S9.1 The Nine Content-Word Phrases

The content-word phrases test whether the effect of *on god* is shared by other two-word phrases that contain a content word. Three of the four two-word controls of the crossed study (*in the*, *with it*, and *at this*) consist of function words, whereas *on god* contains a noun. Each of the nine phrases has two words and contains a content noun or adjective. None contains negation or temporal, locative, or quantity content. Four use an abstract noun or adjective (*in truth*, *in essence*, *in short*, *in reality*), four use a religious noun (*heaven knows*, *lord knows*, *by heaven*, *good heavens*), and one is informal (*for real*). Each phrase encodes to two subwords in all six tokenizers, as *on god* and *in fact* do.

The phrases were screened automatically before inference. The content word of a phrase was already present in at most 0.36% of the pairs of a split. No phrase reached the preregistered exclusion threshold of 5%. The phrases were appended to the hypothesis of every source pair and classified by the six encoders.

The preregistered prediction did not hold. It stated that, on both MultiNLI splits and for each of the five encoders, *on god* would cause at least twice as many failures on clean-correct entailments as each of the nine phrases, with Holm-adjusted $p < .01$ for all nine contrasts. It held in none of the ten encoder-split cells. *By heaven* causes the most failures among the nine phrases in every

one of these cells. The ratio of the failure rate under *on god* to that under *by heaven*, on matched and mismatched, is 0.65 and 0.69 for ELECTRA-small, 0.90 and 0.98 for RoBERTa-large, 1.03 and 1.02 for RoBERTa-base, 0.80 and 0.85 for TimeLM-21, and 0.80 and 0.84 for BERTweet. A second prediction, that no content-word phrase would exceed 7.8% all-five failure on MultiNLI, also did not hold. All-five failure on matched and mismatched is 80.7% and 82.4% for *by heaven*, 24.7% and 24.5% for *good heavens*, 13.9% and 11.9% for *heaven knows*, and 10.8% and 8.8% for *lord knows*. It is at most 7.3% for the other five phrases.

Table S16 gives the paired accuracy change and the flip rate of each phrase. DeBERTa-v3-base orders the phrases differently from the five encoders. All values come from one training run (seed 42).

The content-word phrases were not reviewed for label preservation. Like *on god*, the phrase *by heaven* admits a literal reading that adds content to the hypothesis. Under that reading a neutral label is defensible for some pairs. The accuracy changes in Table S16 therefore show sensitivity and are not established as model errors.

S9.2 Spacing Before the Appended Phrase

The crossed study and the content-word phrases use different spacing rules. The crossed study adds no space before an appended phrase when the hypothesis already ends in whitespace. The content-word phrases were appended after one added space. The two rules differ for 1,291 MultiNLI matched and 1,348 mismatched pairs and for no SNLI test pair. They give different token sequences only for RoBERTa-large and RoBERTa-base.

We measured the effect by rerunning the affected pairs under the rule of the crossed study. Under the original rule the rerun reproduces every saved prediction of all six encoders. Under the rule of the crossed study, the five-encoder mean accuracy changes and flip rates move by at most 0.07 points. The loss under *by heaven* becomes 31.06 points in place of 31.04, and under *good heavens* 14.44 in place of 14.38. All-five failure moves by at most 0.6 points. Under *by heaven* it becomes 81.3% and 82.9% in place of 80.7% and 82.4%. DeBERTa-v3-base values do not change. The order of the nine phrases, the outcome of the prediction, and the comparison with the markers are the same under both rules. The values in this appendix and in the main text are those of the original rule, which the distributed predictions reproduce.

S9.3 Content-Word Coding of All 32 Phrases

We coded all 32 appended phrases by one rule. A phrase is content-bearing if at least one of its words is an English noun, adjective, or lexical verb in its primary dictionary sense. Abbreviations and respellings (*fr*, *frfr*, *tbh*, *ngl*, *istg*, *deadass*), prepositions, determiners, pronouns, the negator *no*, and adverbs do not count. Two alternative codings cover the two judgment calls. The first counts the adverbs *seriously*, *honestly*, *nearby*, and *outside* as content words. The second

counts *deadass* as content-bearing. The outcome is the flip rate, which needs no label. Table S17 lists every phrase with its coding.

Content-bearing phrases change more predictions on average. For the five encoders, the 17 content-bearing phrases have a mean flip rate of 16.30%, against 9.27% for the other 15. The difference is 7.04 points (one-sided permutation test over phrases, $p = .003$, unadjusted). A content-bearing phrase has the higher flip rate in 78.0% of the pairs formed with another phrase. For DeBERTa-v3-base the means are 9.74% and 6.09% ($p = .038$). The two alternative codings give differences of 6.63 points (adverbs) and 8.32 points (*deadass*) for the five encoders, with p at most .005.

The difference is confined to two-word phrases. The 13 content-bearing two-word phrases have a mean five-encoder flip rate of 18.39% (range 6.17 to 39.22). The three two-word phrases without a content word (*in the*, *with it*, *at this*) have 5.20% (4.55 to 5.83). The four single content words (*god*, *talk*, *real*, *cap*) have 9.53% (6.66 to 11.58), against 10.28% (2.51 to 22.44) for the other 12 single words. For DeBERTa-v3-base the four means are 11.20%, 4.09%, 5.00%, and 6.59%.

The coding has exceptions in both directions. Two content-bearing phrases have low flip rates: *in fact* (6.17%) and *in essence* (6.91%). Three phrases coded as not content-bearing have high flip rates: *deadass* (22.44%), *no* (16.16%), and *nearby* (13.29%).

Four caveats limit this analysis. First, the hypothesis and the coding rule were written after the flip rates were known. The alternative codings address the two judgment calls, not the choice of the rule. Second, number of words and content are confounded: 13 of the 16 two-word phrases are content-bearing, and the two-word comparison group has three phrases. Third, the 32 phrases are not a sample of any population, and nine were chosen because they contain content words. Fourth, the phrases overlap and are not independent: *god* is part of *on god*, and three phrases contain *heaven*. The coding separates the groups on average and does not show that content words cause the effect. All values come from one training run (seed 42).

S10 DeBERTa-v3-base in the Crossed Study

DeBERTa-v3-base was evaluated on the same source pairs and the same 24 conditions as the five encoders. Its SNLI and MultiNLI clean baselines were trained under the same protocol. It has one training run (seed 42) and is never pooled with the five encoders. No hypothesis test was computed for DeBERTa-v3-base in the crossed study, so every statement in this section is descriptive.

Ranking of conditions. Among the 23 conditions of the crossed study, *no cap* is the most damaging for DeBERTa-v3-base on all three splits (Table S18). It lowers accuracy against inherited labels by 20.07, 17.84, and 14.86 points on SNLI, MultiNLI matched, and mismatched. The next conditions are a formal phrase and a locative. *Seriously* ranks second on SNLI and third on both MultiNLI

splits. *Nearby* ranks fourth on SNLI and second on both MultiNLI splits. For the five encoders, by contrast, *on god* has the largest mean loss (20.93 points, against 15.99 for *no cap*).

Position of on god. *On god* is not the most damaging condition for DeBERTa-v3-base. It ranks sixth of 23 on SNLI and fourth on both MultiNLI splits, behind *no cap*, *nearby*, and *seriously*. It lowers accuracy by 10.65, 7.84, and 8.63 points on the three splits and changes 14.1%, 10.4%, and 11.6% of predictions. It causes failures on 29.8%, 26.9%, and 30.3% of the clean-correct entailments of DeBERTa-v3-base. On MultiNLI, 97% of these failures go to neutral on each split. The shift to neutral does not distinguish *on god* for this encoder. On MultiNLI, 89% to 100% of its failures go to neutral under every condition of Table S18.

Markers against controls. The margin between the strongest markers and the controls is smaller for DeBERTa-v3-base. We compare the mean accuracy change of *on god*, *deadass*, *no cap*, and *real talk* with the mean over the eight formal and other controls. For DeBERTa-v3-base the markers exceed the controls by 4.51, 4.31, and 3.50 points on SNLI, MultiNLI matched, and mismatched. For the five encoders the margins are 10.99, 12.95, and 13.66 points. On MultiNLI the margin of DeBERTa-v3-base comes almost entirely from *no cap*. The control set includes the two locatives.

Marker-to-control flip-rate ratio. For DeBERTa-v3-base the nine markers change fewer predictions on average than the eight controls. Its mean flip rate over all splits is 6.95% for the markers and 8.00% for the controls, a ratio of 0.868. Each of the five encoders has a ratio above one: 1.329 for ELECTRA-small, 1.657 for RoBERTa-large, 3.08 for RoBERTa-base, 1.994 for TimeLM-21, and 1.603 for BERTweet. The flip rate is sensitivity, not error. For DeBERTa-v3-base, flip rate and accuracy loss agree across the 23 conditions as they do for the five encoders (Spearman $\rho = 0.983$; 11 of 253 condition pairs in inverted order).

Share of the all-five failures. DeBERTa-v3-base shares a minority of the all-five failures under *on god* on MultiNLI. All five encoders fail on 1,048 of 2,518 matched entailments and on 1,215 of 2,544 mismatched entailments. DeBERTa-v3-base also fails on 419 (40.0%) and 499 (41.1%) of these items. On SNLI it fails on 232 of the 263 all-five failures (88.2%). Its failure rate is lower on items that all five encoders classify correctly under *on god*: 22.66% on SNLI, 14.34% on matched, and 16.44% on mismatched. The overlap with the all-five failures is therefore selective, but partial. Measured on the same denominators of 2,518 and 2,544 entailments, the share on which DeBERTa-v3-base and all five encoders fail is 16.6% and 19.6%, against 41.6% and 47.8% for the five encoders alone.

Content-word phrases. The ranking above covers the 23 conditions of the crossed study only. Among the content-word phrases (App. S9), *by heaven* lowers the accuracy of DeBERTa-v3-base by 17.74 points and *heaven knows* by 16.03 points, on average over the three splits, against 17.59 points for *no cap*. On SNLI, *by*

heaven lowers its accuracy by 34.80 points, against 20.07 for *no cap*. On MultiNLI matched and mismatched, *heaven knows* lowers it by 18.25 and 19.26 points, against 17.84 and 14.86 for *no cap*. The content-word phrases were not reviewed for label preservation.

S11 Other Models

This section reports two models that were not trained under the protocol of the paper. Neither enters any table or mean of the main text. Their absolute accuracies are not comparable with those of the six encoders.

S11.1 A Public DeBERTa-v3 NLI Checkpoint

The public checkpoint is MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli, evaluated as released. Its publisher trained it on the MultiNLI training set, FEVER-NLI, and ANLI. The MultiNLI validation splits are therefore in-domain for it, and SNLI is out of domain. We classified every source pair of the three splits under the 24 conditions of the crossed study with the default tokenizer of the checkpoint. No hypothesis test was computed. Its clean accuracy is 88.66% on SNLI test, 90.33% on MultiNLI matched, and 90.28% on mismatched.

Under *on god*, the public checkpoint loses 17.26 points on SNLI, 19.12 on MultiNLI matched, and 21.64 on mismatched (Table S19). *On god* changes 24.5%, 24.2%, and 27.3% of its predictions. It causes failures on 58.4%, 63.9%, and 73.3% of the entailments that the checkpoint classifies correctly on the clean pair. On MultiNLI, 96% and 98% of these failures go to neutral. Accuracy is scored against inherited labels, and the crossed conditions were not reviewed for label preservation.

The checkpoint is sensitive to several appended words, not to *on god* in particular. *No cap* is its most damaging condition on SNLI (28.55 points). The locative *nearby* is its most damaging condition on both MultiNLI splits (25.87 and 26.99 points), where it causes failures on 84.7% and 89.4% of clean-correct entailments. *Outside* lowers accuracy by 15.69, 13.66, and 13.08 points on the three splits. The component word *god* alone accounts for most of the loss under *on god*. For the five encoders, *god* alone is much weaker than the phrase (App. S8).

The checkpoint fails on most items of the all-five failure set, but this reflects its high overall failure rate. Under *on god*, it fails on 254 of the 263 all-five failures on SNLI (96.6%), 825 of 1,048 on MultiNLI matched (78.7%), and 993 of 1,215 on mismatched (81.7%). It also fails on 53.45%, 49.76%, and 62.98% of the items that all five encoders classify correctly under *on god*. The result shows that one public DeBERTa-v3 NLI model is sensitive to appended phrases. It does not show that the effect is specific to *on god*, and it says nothing about DeBERTa-v3-base under the protocol of the paper (App. S10).

S11.2 A Prompted Model

We ran a small test of one prompted model, `gpt-5.6-sol`, without demonstrations and at low reasoning effort. The test covers SNLI only and three conditions: emoji substitution, a held-out marker of fold 1 at the end of the hypothesis, and the combined condition. For each condition, 166 edited SNLI test pairs were drawn by a deterministic sampler that approximately balances the labels (56 entailment, 55 neutral, 55 contradiction). The selection did not use model predictions. Each edited pair was classified together with its paired clean pair. The prompt defines the three labels and asks the model to interpret the text as written, including emoji, informal language, and discourse markers. It therefore directs attention to the edits, which the fine-tuned encoders never receive. The model received only the premise and the hypothesis of each pair. One prompt was used. The test per condition is the two-sided exact McNemar test, unadjusted.

The test shows no significant paired difference in any condition (Table S20). The differences are +2.41, +1.20, and 0.00 points, with $p = .289, .688, \text{ and } 1.0$. Aggregate accuracy can stay the same while predictions change. In the combined condition, eight pairs become correct and eight become incorrect.

This test supports no robustness claim. It does not show that the model is insensitive to the edits, and it establishes neither equivalence nor superiority. The sample has 166 pairs per condition and 6 to 16 discordant pairs, so a loss of a few points could go undetected. The test covers one split, one prompt, and the held-out markers of one fold. It includes none of the conditions of the crossed study, and so not *on god*. The training data of the model, and its possible exposure to SNLI, are unknown.

S12 The Two Fixes Together

This section covers the five encoders that have augmentation and hybrid systems. DeBERTa-v3-base has neither. Accuracies are against inherited labels, and magnitudes are from one training run (seed 42) unless stated otherwise.

S12.1 Marker Augmentation Transfer

Marker augmentation raises accuracy on held-out markers in every fold and for every encoder, by less than on seen markers on average (Table S21). This contrast is exploratory. Each gain is the accuracy of the augmented checkpoint minus that of the clean baseline on the same marked input. The held-out gain is positive in 43 of 45 cells. It is significantly positive in 42 cells and significantly negative in one (ELECTRA-small, MultiNLI matched, fold 3: -1.26 pp). Two cells are not significant. The seen gain is significantly positive in all 45 cells. The tests are McNemar tests, Holm-adjusted within a family of 930 exploratory tests per encoder, split, and fold.

The share recovered differs more across encoders than across folds. It is 33.1% for ELECTRA-small and 98.6% for TimeLM-21. The share for fold 2 rests on a small loss and is unstable. The separately trained hybrid checkpoints gain 7.47 points on held-out markers.

S12.2 Clean-Accuracy Cost of the Hybrid

For ELECTRA-small, the hybrid is non-inferior to the clean baseline on clean input in all nine split-fold cells (Table S22). The margin of 0.5 points was prespecified, and the five training runs follow a preregistered design. For each cell, the test uses the mean over the five runs of the difference in clean accuracy, hybrid minus baseline. The hybrid is non-inferior if the one-sided 95% lower bound of this mean exceeds -0.5 points. The bound is from Student’s t with four degrees of freedom. The bounds range from -0.35 to -0.11 points. The nine cells are tested separately, with no adjustment across cells. The test does not show equivalence.

For the other four encoders the differences are descriptive. Each has three runs in fold 1, and no test was planned. The mean differences range from -0.24 to $+0.26$ points. The intervals are as wide as -1.17 to $+1.29$ points, too wide to support non-inferiority at this margin.

S12.3 Interaction of the Two Fixes

We find no evidence that emoji normalization and marker augmentation interact on combined input. The analysis is exploratory. It covers ELECTRA-small and TimeLM-21 in fold 1, which gives six encoder-split cells. The interaction is the accuracy of the hybrid, minus that of augmentation, minus that of normalization, plus that of the clean baseline. We subtract the same quantity computed on the paired clean pairs. The result ranges from -0.77 to $+1.92$ points, and none of the six cells is significant after Holm correction. Without the subtraction the range is $+0.17$ to $+1.39$ points, and again no cell is significant. The 95% intervals on MultiNLI span about three points on either side, so an interaction of that size is not excluded. The augmented and hybrid checkpoints were trained separately, so the estimate is not factorial.

S12.4 Composition of the Two Edits

On combined input, the clean baseline fails on nearly the same share of pairs as fail under at least one edit alone (Table S23). The analysis is exploratory. It is restricted to pairs that the baseline classifies correctly in their paired clean form. Pooled over splits and folds, the failure rate on combined input differs from the union of the two single-edit failure rates by -0.31 to $+0.50$ points per encoder. Across the 45 cells the difference ranges from -1.74 to $+2.44$ points. One cell is an exception: RoBERTa-large on SNLI in fold 2 ($+1.02$ points, Holm-adjusted $p = 1.05 \times 10^{-4}$, exact McNemar test). No other cell is significant after correction. The pooled rows count the same source pairs in several folds and are descriptive.

S12.5 Emoji Restoration on the Same Checkpoint

Restoring the replaced nouns in combined input raises accuracy by about four points, with or without marker augmentation. The analysis is exploratory, and

no test is attached to it. Restoration undoes the recorded emoji spans and leaves the marker in place. The mean gain over three splits and three folds is 3.96 points for the clean baseline (77.67% to 81.63%) and 4.06 points for the augmented checkpoint (80.75% to 84.81%). Per encoder the gain is 3.72 to 4.41 points and 3.38 to 5.05 points.

The comparison has three limits. First, the restored text is identical to the held-out marker condition of the same source in all 22,578 cases. The gain therefore equals the emoji loss in the presence of a marker and is not an independent test of a fix. Second, the two checkpoints are separate training runs, so the comparison between them is not factorial. Third, values for single cells are not available, so we make no statement about individual cells.

S13 Case Study of *on god*

On god was chosen after its effect was seen on the same evaluation data, so no analysis here confirms that effect independently. A failure is a prediction other than the inherited label. No pair with an appended phrase was independently labeled, and *on god* admits a literal reading. Failure rates therefore show sensitivity and not established error. All results use one training run (seed 42). DeBERTa-v3-base is a separate row and is not part of any all-five quantity.

S13.1 All-Five Failure and Its Destination

All-five failure under *on god* exceeds the count expected from the failure rates of the single encoders (Table S24). The expected count assumes that each encoder’s failures fall on random items. This comparison is exploratory. The ratio of observed to expected counts is 1.57 and 1.28 on MultiNLI. It is 10.46 on SNLI, where single-encoder failure rates are lower. The expected count does not remove the dependence among the five checkpoints, which share training data, and it does not model item difficulty.

On SNLI, *on god* does not lead the other conditions. All-five failure is 20.30% under *deadass*, 13.82% under *seriously*, and 12.64% under *outside*. *By heaven* produces all-five failure of 78.0%, 80.7%, and 82.4% on the three splits. Expected counts were not computed for it, and it was not reviewed for label preservation, so these rates are descriptive.

On MultiNLI, the failures of every encoder go mostly to neutral (Table S25). The share is 89.08% to 98.96%. On SNLI the destination differs by encoder. The share to neutral is 40.65% for ELECTRA-small, 74.64% for BERTweet, 76.54% for RoBERTa-large, and 83.37% for TimeLM-21. RoBERTa-base sends 2,214 of its 2,334 failures to contradiction.

S13.2 High-Agreement Items from ChaosNLI

All-five failure under *on god* persists on items with high-agreement labels, at a lower rate than on the other items. The check is exploratory and covers MultiNLI

matched only. ChaosNLI provides 100 labels for each of 1,599 matched pairs. Of these, 195 received at least 80 entailment votes, 160 of the 195 carry the inherited label entailment, and 83 of the 160 are clean-correct for all five encoders. On these 83 items, all five encoders fail under *on god* on 26 (31.3%, Wilson 95% interval 22.4–41.9). Under *no cap*, the comparison phrase, they fail on 5 (6.0%). Among items with fewer than 80 entailment votes, the rate is 45 of 107 (42.1%, Wilson 95% interval 33.1–51.6). The two intervals overlap.

The check has two limits. The votes concern the pair without the suffix, and no annotator saw the hypothesis with *on god*. The threshold allows up to 20 dissenting votes, and every ChaosNLI item had a disputed original label (five votes split 3–2 for 1,302 items and 3–1–1 for 297). The check therefore does not exclude that ambiguity contributes to the failures.

S13.3 Transfer to SICK and ANLI

The failures under *on god* transfer to SICK and ANLI, but the specificity criterion failed on both (Table S26). The analysis was preregistered. The MultiNLI-trained checkpoints were applied to the SICK test set (4,906 pairs) and the ANLI development sets (3,200 pairs). The controls were *in fact*, *nearby*, *seriously*, *honestly*, and *in the*. The specificity criterion required, for each of the five encoders, a failure rate under *on god* of at least three times that of the strongest control, with Holm-adjusted $p < .01$ against each control. It counted as failed for a dataset if it held for at most two encoders. Three further predictions were that at least 80% of failures go to neutral, that all-five failure is at least three times that of every control, and that the ratio for DeBERTa-v3-base is below 2.

The criterion held for none of the five encoders on either dataset, because the controls also cause many failures. The ratio is 1.16 to 2.70 on SICK and 0.99 to 1.92 on ANLI. On SICK, *honestly* or *seriously* causes failures on 60.4% to 83.3% of clean-correct entailments for four of the five encoders. The other three predictions held, except that the share to neutral on ANLI is 69% for ELECTRA-small. On SICK, all-five failure under *on god* is 80.0% (905 of 1,131 items), against 15.5% under *honestly*, the strongest control. Expected counts were not computed for SICK or ANLI. Each encoder fails on 88.9% to 99.1% of its own items on SICK, so the shared rate adds little evidence.

The ANLI results carry little weight. Clean accuracy is 27.59% to 31.66% for the five encoders and 37.00% for DeBERTa-v3-base, near chance for three labels. The paired accuracy change under *on god* is between -1.72 and $+2.19$ points. Only 45 entailments are clean-correct for all five encoders. All five fail on 10 of them under *on god* and on none under each control.

SNLI-trained checkpoints show no specificity on SICK. Their clean accuracy on SICK is 55.06% to 58.13% for the five encoders. *Seriously* is the strongest control for every encoder. The ratio of *on god* to *seriously* is 0.78 to 1.11 for the five encoders and 0.82 for DeBERTa-v3-base. All-five failure is 63.0% under *seriously* and 45.4% under *on god* (1,159 items).

S13.4 The Training-Data Account

A preregistered account based on the training data met none of its four criteria (Table S27). The account proposed that MultiNLI training hypotheses containing *god* are labeled neutral more often than others, so that the word is a learned cue for neutral, and that SNLI lacks this cue. The decision rule required at least three of the four criteria. The measurements are observational. They do not intervene on the training data or on the encoders. A skew that is not significant does not show that no cue was learned. The cause of the failures is not identified.

S13.5 Contrast with Other Commitment Markers

On MultiNLI, encoders respond differently to markers usually taken to share a commitment function (Table S25). The analysis is exploratory, and the shared function is an assumption. For RoBERTa-base, RoBERTa-large, and BERTweet, *on god* causes failures on 72.99% to 97.10% of clean-correct entailments. *Honestly* and *seriously* cause failures on 3.83% to 21.48% of the same items. The contrast is weaker for ELECTRA-small and TimeLM-21. For these two, the rate under *on god* is 1.52 to 1.59 times that under *honestly*. For DeBERTa-v3-base, *on god* and *seriously* do not differ significantly ($p = .063$ and $p = .97$, exact McNemar test, unadjusted). This is a failure to reject and not evidence of equal rates.

The contrast is absent on SICK, where *honestly* or *seriously* causes failures on most clean-correct entailments for four of the five encoders (Table S26). The contrast concerns the encoders only. It does not show how annotators would label the pairs with the suffix. *By heaven*, an oath, causes failures on 91.6% to 97.0% of the matched clean-correct entailments of the five encoders. A shared commitment function therefore does not separate the phrases with large losses from the others.

Table S6. Test families. McNemar (cc): chi-square McNemar test with continuity correction. McNemar (exact): two-sided exact binomial McNemar test. t bound: one-sided Student- t lower bound on the mean difference. Holm correction is applied within each family at $\alpha = .05$ and never across families.

What is compared	Test	Tests per family; correction	Status
Four specified contrasts, five encoders: emoji loss, marker loss, normalization versus augmentation, hybrid versus baseline	McNemar (cc)	4 per encoder, split, and fold (45 families); Holm	specified before training
Fold study: marker versus formal phrase and versus other appended word, five encoders	McNemar (cc)	36 per encoder and split (15 families, 540 tests); Holm	family fixed before the run; comparisons exploratory
Gain from marker augmentation, five encoders	McNemar (cc)	930 per encoder, split, and fold; Holm	exploratory
Crossed study, five encoders: marker versus component word (90 failure and 30 neutral-prediction contrasts); all-five failure, <i>on god</i> versus four two-word controls (12)	McNemar (exact)	132 in one family; Holm	fixed before the run
DeBERTa-v3-base: emoji loss (1), marker loss per fold (3), marker versus control (6), normalization (1)	McNemar (exact)	11 per split (3 families); Holm	not preregistered
Content-word phrases: failures under <i>on god</i> versus each of nine phrases, six encoders	McNemar (exact)	9 per encoder and split (18 families); Holm	preregistered
SICK and ANLI: failures under <i>on god</i> versus each of ten other conditions, six encoders	McNemar (exact)	10 per encoder and dataset (12 families); Holm	preregistered
Clean accuracy of hybrid versus clean baseline, ELECTRA-small; non-inferiority margin 0.5 pp	one-sided 95% t bound, five runs	9 split-fold cells, tested separately; none	margin specified before training; runs preregistered

Table S7. Agreement of the additional runs with seed 42, by contrast. Entries are counts of cells. Interval: 95% Student-*t* interval across runs. Range: range of the additional runs. N–A: normalization minus augmentation on emoji input. H–B: hybrid minus baseline on combined input. Five encoders; DeBERTa-v3-base is not covered.

	Emoji loss	Marker loss	N–A	H–B	Total
Cells	15	21	21	21	78
All runs in the expected direction	15	21	21	21	78
Interval excludes zero	15	21	18	20	74
Seed 42 outside the range	8	11	11	9	39

Table S8. Paired accuracy change (pp) under emoji substitution by replacement structure, SNLI test, seed 42, against inherited labels. The five-encoder means are in the main text.

Replacement structure	Pairs	EL	RB	RL	TL	BT	DeB
One noun, both sentences	2,934	−0.51	−0.44	−0.55	−0.51	−0.78	−0.48
One noun, premise only	1,282	−6.40	−3.59	−3.90	−5.38	−4.84	−0.62
One noun, hypothesis only	424	−8.02	−6.84	−12.74	−17.22	−4.72	−3.77
Two nouns, same sentence	1,301	−17.68	−11.91	−11.61	−14.53	−14.45	−2.08
Two nouns, one per sentence	489	−23.52	−20.86	−20.65	−18.00	−21.06	−8.18

Table S9. Paired accuracy change (pp) under emoji substitution for the three one-noun structures on the two MultiNLI validation splits (seed 42, against inherited labels). Mean: mean of the five encoders EL to BT.

Replacement structure	Pairs	Mean	EL	RB	RL	TL	BT	DeB
<i>MultiNLI matched</i>								
One noun, both sentences	152	−0.79	−1.32	+0.66	−1.32	−1.97	0.00	−1.97
One noun, premise only	204	−0.78	−2.45	0.00	−0.98	−0.98	+0.49	+0.49
One noun, hypothesis only	140	−7.71	+1.43	−10.00	−8.57	−11.43	−10.00	−6.43
<i>MultiNLI mismatched</i>								
One noun, both sentences	141	−1.28	0.00	−0.71	−2.13	−2.13	−1.42	−0.71
One noun, premise only	254	−1.34	−1.57	−1.97	−1.97	−0.39	−0.79	−2.76
One noun, hypothesis only	122	−6.56	−7.38	−13.93	−3.28	−8.20	0.00	−7.38

Table S10. Paired accuracy change (pp) by replaced noun, SNLI test pairs with one replaced noun (seed 42, against inherited labels). Mean: mean of the five encoders. DeB: DeBERTa-v3-base. Nouns are ordered by number of pairs.

Noun	Pairs	Mean	DeB	Noun	Pairs	Mean	DeB
<i>man</i>	1,895	-1.92	-0.37	<i>camera</i>	69	0.00	+1.45
<i>woman</i>	865	-3.70	-0.23	<i>baby</i>	33	-1.21	-3.03
<i>boy</i>	414	-1.74	+0.24	<i>horse</i>	28	-1.43	-3.57
<i>dog</i>	363	-2.81	-1.10	<i>hat</i>	26	+1.54	-7.69
<i>girl</i>	312	-3.33	-2.88	<i>bird</i>	21	-11.43	-4.76
<i>person</i>	300	-3.73	-1.00	<i>guitar</i>	19	+2.11	0.00
<i>child</i>	197	-3.86	-2.54	<i>bicycle</i>	16	-23.75	-25.00
<i>car</i>	77	+1.82	-1.30	<i>cat</i>	5	-8.00	0.00

Table S11. Permutation p values for equal mean paired accuracy change across levels, SNLI test (2,000 permutations, unadjusted, seed 42).

Encoder	Across the five structures (6,430 pairs)	Across nouns, one replaced noun (4,640 pairs)	Across nouns, one noun, both sentences (2,934 pairs)
ELECTRA-small	0.0005	0.0015	0.1999
RoBERTa-base	0.0005	0.007	0.008
RoBERTa-large	0.0005	0.0005	0.1039
TimeLM-21	0.0005	0.0005	0.0695
BERTweet	0.0005	0.003	0.3648
DeBERTa-v3-base	0.0005	0.001	0.2884

Table S12. Held-out marker minus control, in pp of accuracy against inherited labels, at the end of the hypothesis (five encoders, seed 42). Negative values mean that the marker is more damaging than the control. Cells: number of model-split cells in which the control is more damaging than the marker.

Fold	Held-out markers	Formal phrases		Other appended words	
		Difference	Cells	Difference	Cells
1	<i>frfr, deadass, no cap</i>	-8.45	0/15	-7.66	0/15
2	<i>fr, tbh, ngl</i>	+2.31	10/15	+3.48	13/15
3	<i>real talk, on god, istg</i>	-9.10	0/15	-8.80	1/15
All		-5.08	10/45	-4.33	14/45

Table S13. Change in accuracy against inherited labels (pp) for held-out markers by position, clean baselines, seed 42. Means over three splits and three folds.

Encoder	End of hypothesis	Start of hypothesis	End of premise
ELECTRA-small	−12.83	−7.57	−3.08
RoBERTa-base	−11.08	−6.95	−0.21
RoBERTa-large	−9.71	−4.96	−0.18
TimeLM-21	−11.20	−5.87	−0.10
BERTweet	−5.85	−2.24	−0.20
Mean of five	−10.13	−5.52	−0.75

Table S14. The 23 non-clean conditions of the crossed study (seed 42). Paired accuracy change against inherited labels (Acc., pp) and flip rate (Flip, %). Five encoders: mean over 15 model–split cells. DeBERTa-v3-base: mean over three splits, one training run, reported separately. (loc.): locative adverb.

Condition	Five encoders		DeBERTa-v3-base	
	Acc.	Flip	Acc.	Flip
<i>Markers</i>				
<i>on god</i>	−20.93	27.3	−9.04	12.0
<i>deadass</i>	−16.90	22.4	−8.03	11.2
<i>no cap</i>	−15.99	22.1	−17.59	21.8
<i>real talk</i>	−14.53	19.9	−4.13	6.4
<i>frfr</i>	−7.53	11.2	−2.14	3.1
<i>ngl</i>	−5.55	9.0	−0.91	1.6
<i>tbh</i>	−4.15	7.7	−0.36	1.6
<i>istg</i>	−3.03	5.8	−1.17	2.3
<i>fr</i>	−2.59	4.8	−1.45	2.4
<i>Component words</i>				
<i>no</i>	−10.68	16.2	−4.55	7.0
<i>god</i>	−7.45	11.6	−3.10	5.1
<i>talk</i>	−6.69	10.0	−2.62	4.3
<i>real</i>	−6.44	9.9	−4.83	7.2
<i>cap</i>	−4.01	6.7	−2.01	3.3
<i>on</i>	−0.57	2.5	−1.87	3.1
<i>Formal phrases</i>				
<i>seriously</i>	−6.69	10.3	−11.72	14.9
<i>honestly</i>	−6.26	9.8	−4.03	6.3
<i>in fact</i>	−2.22	6.2	−2.70	5.2
<i>Other appended words</i>				
<i>nearby</i> (loc.)	−8.78	13.3	−11.56	15.2
<i>outside</i> (loc.)	−6.48	10.4	−7.51	10.2
<i>in the</i>	−2.89	5.8	−4.24	6.1
<i>with it</i>	−1.79	5.2	−1.87	3.6
<i>at this</i>	−1.30	4.5	−1.10	2.6

Table S15. Component-word contrasts on clean-correct entailments: number of model-split cells (of 15, five encoders) in which the marker causes significantly more failures than the component word, significantly fewer, or neither (Holm-adjusted, $\alpha = .05$; seed 42).

Comparison	More	Fewer	Neither
<i>on god</i> vs. <i>on</i>	15	0	0
<i>on god</i> vs. <i>god</i>	13	2	0
<i>real talk</i> vs. <i>real</i>	14	1	0
<i>real talk</i> vs. <i>talk</i>	12	2	1
<i>no cap</i> vs. <i>no</i>	11	4	0
<i>no cap</i> vs. <i>cap</i>	14	0	1
Total	79	9	2

Table S16. The nine content-word phrases, with *on god* for reference (seed 42). Paired accuracy change against inherited labels (Acc., pp) and flip rate (Flip, %). Five encoders: mean over 15 model-split cells. DeBERTa-v3-base: mean over three splits, reported separately. The phrases were not reviewed for label preservation.

Phrase	Five encoders		DeBERTa-v3-base	
	Acc.	Flip	Acc.	Flip
<i>by heaven</i>	−31.04	39.22	−17.74	21.67
<i>good heavens</i>	−14.38	19.90	−7.36	10.17
<i>heaven knows</i>	−14.03	19.83	−16.03	20.07
<i>in reality</i>	−12.89	18.69	−4.83	7.89
<i>lord knows</i>	−12.19	17.34	−10.54	13.69
<i>in truth</i>	−10.17	15.42	−3.39	5.86
<i>for real</i>	−10.11	15.19	−8.84	12.15
<i>in short</i>	−6.57	11.03	−3.29	5.02
<i>in essence</i>	−2.73	6.91	−1.53	3.71
<i>on god</i>	−20.93	27.28	−9.04	12.02

Table S17. The 32 appended phrases in order of five-encoder mean flip rate (Flip, %; seed 42). Family: M marker, W component word, F formal phrase, O other appended word, C content-word phrase. Cod.: content-bearing under the main coding rule. ^aContent-bearing when adverbs count. ^bContent-bearing in the second alternative coding. (loc.): locative adverb.

Phrase	Fam.	Cod.	Flip	Phrase	Fam.	Cod.	Flip
<i>by heaven</i>	C	yes	39.22	<i>outside</i> (loc.)	O	no ^a	10.38
<i>on god</i>	M	yes	27.28	<i>seriously</i>	F	no ^a	10.28
<i>deadass</i>	M	no ^b	22.44	<i>talk</i>	W	yes	9.96
<i>no cap</i>	M	yes	22.11	<i>real</i>	W	yes	9.91
<i>real talk</i>	M	yes	19.94	<i>honestly</i>	F	no ^a	9.81
<i>good heavens</i>	C	yes	19.90	<i>ngl</i>	M	no	9.05
<i>heaven knows</i>	C	yes	19.83	<i>tbh</i>	M	no	7.69
<i>in reality</i>	C	yes	18.69	<i>in essence</i>	C	yes	6.91
<i>lord knows</i>	C	yes	17.34	<i>cap</i>	W	yes	6.66
<i>no</i>	W	no	16.16	<i>in fact</i>	F	yes	6.17
<i>in truth</i>	C	yes	15.42	<i>in the</i>	O	no	5.83
<i>for real</i>	C	yes	15.19	<i>istg</i>	M	no	5.76
<i>nearby</i> (loc.)	O	no ^a	13.29	<i>with it</i>	O	no	5.23
<i>god</i>	W	yes	11.58	<i>fr</i>	M	no	4.83
<i>frfr</i>	M	no	11.16	<i>at this</i>	O	no	4.55
<i>in short</i>	C	yes	11.03	<i>on</i>	W	no	2.51

Table S18. DeBERTa-v3-base in the crossed study (one training run, seed 42): paired accuracy change against inherited labels (pp) and rank among the 23 conditions by accuracy loss, per split. Selected conditions. (loc.): locative adverb.

Condition	Family	SNLI		MultiNLI m		MultiNLI mm	
		Acc.	Rank	Acc.	Rank	Acc.	Rank
<i>no cap</i>	Marker	-20.07	1	-17.84	1	-14.86	1
<i>on god</i>	Marker	-10.65	6	-7.84	4	-8.63	4
<i>deadass</i>	Marker	-13.89	3	-5.93	6	-4.27	6
<i>real talk</i>	Marker	-7.50	10	-2.44	9	-2.44	9
<i>god</i>	Component	-1.54	22	-3.81	7	-3.95	7
<i>on</i>	Component	-3.99	16	-0.87	16	-0.75	16
<i>seriously</i>	Formal	-18.28	2	-7.96	3	-8.92	3
<i>honestly</i>	Formal	-6.71	11	-2.87	8	-2.51	8
<i>in fact</i>	Formal	-4.06	15	-2.21	11	-1.83	12
<i>nearby</i> (loc.)	Other	-12.93	4	-11.14	2	-10.62	2
<i>outside</i> (loc.)	Other	-11.51	5	-5.95	5	-5.07	5

Table S19. Public DeBERTa-v3 NLI checkpoint, not trained under the protocol of the paper: paired accuracy change against inherited labels (pp) for selected conditions of the crossed study. Flip: flip rate under *on god* (%). (loc.): locative adverb.

Split	<i>on god</i>		<i>god</i>	<i>on</i>	<i>no cap</i>	<i>seriously</i>	<i>in fact</i>	<i>nearby</i> (loc.)
	Acc.	Flip						
SNLI	-17.26	24.5	-13.59	-1.06	-28.55	-15.71	-0.83	-17.20
MultiNLI m	-19.12	24.2	-14.11	-1.17	-20.02	-11.43	-1.84	-25.87
MultiNLI mm	-21.64	27.3	-17.67	-1.17	-17.40	-13.90	-2.14	-26.99

Table S20. Prompted model on 166 SNLI test pairs per condition. Accuracy (%) against inherited labels on the paired clean and the edited pairs, their difference (pp), the share of pairs with the same prediction (Same, %), the discordant counts (correct on the clean pair only; correct on the edited pair only), and the unadjusted exact McNemar p .

Condition	Clean	Edited	Diff.	Same	Clean only	Edited only	p
Emoji substitution	90.36	92.77	+2.41	95.18	2	6	.289
Held-out marker	89.16	90.36	+1.20	96.39	2	4	.688
Combined	87.95	87.95	0.00	90.36	8	8	1.0

Table S21. Marker augmentation by encoder and by fold (seed 42, pp). Loss: paired accuracy change of the clean baseline on held-out markers. Gain: augmented checkpoint minus clean baseline. Range: smallest and largest held-out gain over cells. Share: mean held-out gain as a percentage of the mean loss. Encoder rows average nine cells, fold rows 15.

	Loss	Seen gain	Held-out gain	Range	Share (%)
ELECTRA-small	-12.83	+12.60	+4.24	-1.26 to +9.11	33.1
RoBERTa-base	-11.08	+11.34	+8.99	+1.02 to +17.39	81.1
RoBERTa-large	-9.71	+9.51	+7.08	+1.65 to +13.72	72.9
TimeLM-21	-11.20	+11.26	+11.05	+8.28 to +14.07	98.6
BERTweet	-5.85	+5.71	+5.20	+1.04 to +11.65	88.8
Mean of five	-10.13	+10.08	+7.31	-1.26 to +17.39	72.1
Fold 1	-13.47	+8.48	+9.65	—	71.6
Fold 2	-4.16	+12.96	+4.02	—	96.5
Fold 3	-12.77	+8.80	+8.27	—	64.7

Table S22. Clean accuracy of the hybrid minus the clean baseline (pp). Top: ELECTRA-small, mean of five runs, with the one-sided 95% lower bound in parentheses. Bottom: the other four encoders in fold 1, mean of three runs, with the two-sided 95% interval in brackets.

ELECTRA-small	Fold 1	Fold 2	Fold 3
SNLI test	+0.01 (−0.11)	−0.10 (−0.23)	−0.01 (−0.14)
MultiNLI matched	−0.15 (−0.19)	−0.11 (−0.27)	−0.06 (−0.15)
MultiNLI mismatched	+0.02 (−0.23)	−0.02 (−0.26)	−0.08 (−0.35)
Fold 1 only	SNLI test	matched	mismatched
RoBERTa-base	+0.17 [−0.24, 0.57]	+0.26 [−0.17, 0.70]	0.00 [−0.28, 0.28]
RoBERTa-large	−0.04 [−0.40, 0.32]	−0.24 [−0.88, 0.39]	−0.22 [−0.66, 0.21]
TimeLM-21	+0.07 [0.00, 0.14]	−0.01 [−0.74, 0.73]	+0.10 [−0.31, 0.50]
BERTweet	−0.05 [−0.78, 0.69]	−0.22 [−0.91, 0.46]	+0.06 [−1.17, 1.29]

Table S23. Failures of the clean baseline, pooled over splits and folds (seed 42; % of clean-correct pairs), under emoji substitution, a held-out marker, and both. Diff.: failure rate on combined input minus the union of the two single-edit failure rates (pp).

Encoder	Pairs	Emoji	Marker	Combined	Diff.
ELECTRA-small	19,779	9.34	12.10	19.97	+0.40
RoBERTa-base	20,436	7.22	7.84	14.01	+0.15
RoBERTa-large	20,802	7.04	7.42	13.61	+0.37
TimeLM-21	20,442	8.41	8.78	15.81	+0.50
BERTweet	20,205	7.93	3.92	10.66	−0.31

Table S24. All-five failure under *on god* (seed 42). Items: entailments that are clean-correct for all five encoders. Expected: count if each encoder’s failures fell on random items. O/E: observed divided by expected, with a 95% bootstrap interval over source pairs.

Split	Items	Observed	Rate (%)	Expected	O/E
SNLI test	2,887	263	9.11	25.14	10.46 [9.27, 11.79]
MultiNLI matched	2,518	1,048	41.62	668.51	1.57 [1.50, 1.64]
MultiNLI mismatched	2,544	1,215	47.76	949.37	1.28 [1.24, 1.32]

Table S25. *On god* on MultiNLI by encoder (seed 42). Items: the encoder’s clean-correct entailments. Fail: failure under *on god* (%). Neut.: share of these failures predicted neutral (%). Hon., Ser.: failure under *honestly* and *seriously* on the same items (%).

Encoder	MultiNLI matched					MultiNLI mismatched				
	Items	Fail	Neut.	Hon.	Ser.	Items	Fail	Neut.	Hon.	Ser.
ELECTRA-small	2,886	62.82	89.08	39.54	32.71	2,901	68.25	90.86	44.81	30.68
RoBERTa-base	2,998	95.73	98.54	5.37	8.91	3,030	97.10	98.57	5.78	7.33
RoBERTa-large	3,117	83.64	98.96	4.78	6.19	3,135	89.00	98.85	3.83	4.72
TimeLM-21	2,989	77.35	97.66	50.02	12.28	2,987	83.23	98.47	54.70	11.99
BERTweet	2,910	72.99	97.50	21.48	17.59	2,957	79.00	98.12	19.68	17.15
DeBERTa-v3-base	3,036	26.88	97.4	11.20	28.59	3,085	30.34	96.9	9.76	30.41

Table S26. MultiNLI-trained checkpoints on SICK and ANLI (seed 42). Clean: accuracy on unedited pairs (%). Fail and Control: failure (%) on the encoder’s clean-correct entailments under *on god* and under the strongest control (H: *honestly*, S: *seriously*, F: *in fact*). Ratio: Fail divided by Control.

Encoder	SICK test				ANLI development			
	Clean	Fail	Control	Ratio	Clean	Fail	Control	Ratio
ELECTRA-small	53.06	98.9	H 65.9	1.50	27.59	71.3	H 71.8	0.99
RoBERTa-base	56.01	99.1	S 70.1	1.42	28.97	81.1	F 42.3	1.92
RoBERTa-large	56.93	97.0	S 83.3	1.16	31.66	46.7	F 29.5	1.58
TimeLM-21	55.30	88.9	H 60.4	1.47	30.53	90.5	H 65.9	1.37
BERTweet	54.06	90.0	H 33.3	2.70	29.84	85.3	H 59.7	1.43
DeBERTa-v3-base	57.28	90.1	S 87.4	1.03	37.00	57.7	S 57.1	1.01

Table S27. The four preregistered criteria of the training-data account and their outcomes. The classifier is a logistic regression that sees only the hypothesis.

Criterion	Outcome
1. <i>God</i> is skewed to neutral in MultiNLI training hypotheses and nearly absent from SNLI.	Not met. 378 MultiNLI hypotheses, 36.5% neutral ($p = .37$); 36 SNLI hypotheses.
2. The classifier moves more entailments to neutral under <i>on god</i> or <i>god</i> than under two-word controls, and not on SNLI.	Not met. <i>On god</i> exceeds <i>in fact</i> by 4.14 points on matched [2.53, 5.80] and 1.83 on mismatched [0.00, 3.73].
3. The five encoders fail most in the genres where <i>god</i> is frequent, and DeBERTa-v3-base varies less.	Not met. All-five failure on matched is highest for government and lowest for telephone. DeBERTa-v3-base also varies by genre.
4. The controls <i>nearby</i> , <i>seriously</i> , and <i>cap</i> show no such skew.	Not met. <i>Nearby</i> is skewed to neutral (291 hypotheses, 41%, $p = .011$).